Honors Geometry – 3.5 Practice

Full Name: _____

Equations of Parallel and Perpendicular Lines • Section 3.5 Period: _____

Show your work. The work is the point, not the answer. Each part ends with a self-check, so use it before you move on.

Part A — Slope first.

Label (x_1, y_1) and (x_2, y_2), then subtract in the SAME order top and bottom. Ex: (1, −2), (5, 6) → m = (6 − (−2)) / (5 − 1) = 8/4 = 2

1. Find the slope of the line through (−3, 8) and (5, −4).

m = (____ − ____) / (____ − ____) = _____

answer: _____

2. Find the slope of the line through (2, −7) and (2, 5).

m = (____ − ____) / (____ − ____) = _____

answer: _____

3. Find the slope of the line through (−6, 4) and (3, 4).

m = (____ − ____) / (____ − ____) = _____

answer: _____

4. A line with slope −2/5 passes through (−1, 6) and (9, ____). Find the missing coordinate.

−2/5 = (____ − 6) / (9 − (−1))

answer: _____

✓ **Check yourself:** Your four answers: one negative fraction, one undefined, one zero, one positive whole number. Two of anything? #2 and #3 are the usual suspects.

Part B — Parallel, perpendicular, coinciding, or neither?

Solve for y first. Same slope, different b → parallel. Same slope, same b → coinciding. Slopes multiply to −1 → perpendicular. Otherwise → neither. Vertical and horizontal lines are always perpendicular.

5. $y = \frac{3}{4}x - 2$ and $y = \frac{3}{4}x + 5$

m_1 = ____ m_2 = ____

answer: _____

6. $y = -5x + 1$ and $x - 5y = 15$

rewrite: y = _____

answer: _____

7. $2x + 3y = 12$ and $y = -\frac{2}{3}x + 4$

rewrite: y = _____

answer: _____

8. $y = 2x + 7$ and $y = -2x + 7$

m_1 = ____ m_2 = ____

answer: _____

9. $x = -4$ and $y = 9$

vertical or horizontal? _____ _____

answer: _____

10. $6x - 2y = 8$ and $y = 3x + 1$

rewrite: y = _____

answer: _____

✓ **Check yourself:** Exactly two parallel, two perpendicular, one coinciding, one neither. If the tally is off, recheck the problem you did NOT rewrite into y = mx + b.

Part C — Write the equation.

Get the slope of the given line, adjust it (same for parallel, negative reciprocal for perpendicular), push it through the point, solve for y. Ex: through (3, 1) ∥ y = 2x − 9 → y − 1 = 2(x − 3) → y = 2x − 5

11. Through (4, −3), parallel to $y = \frac{1}{2}x + 6$.

answer: _____

12. Through (−6, 1), perpendicular to $y = 3x - 8$.

answer: _____

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13. Through (2, 9), parallel to $4x + y = 5$.

answer: _____

14. Through (−3, −2), perpendicular to $2x − 5y = 10$.

answer: _____

15. Through (0, −7), parallel to $y = −x + 4$.

answer: _____

16. Through (5, 1), perpendicular to $y = \frac{4}{5}x − 3$.

answer: _____

Answer Bank

$y = −4x + 17$ **(O)**	$y = \frac{1}{2}x + 5$ **(T)**
$y = \frac{1}{2}x − 5$ **(S)**	$y = −x − 7$ **(E)**
$y = −3x − 1$ **(A)**	$y = −5/4\ x + 29/4$ **(S)**
$y = −\frac{1}{3}x − 1$ **(L)**	$y = −4x + 9$ **(R)**
$y = −5/2\ x − 19/2$ **(P)**	

✓ **Check yourself:** Write the letter for #11 – 16 in order. They spell a word this section is all about. Three entries are decoys: land on one and that problem is wrong.

11	12	13	14	15	16

Part D — Push further.

Same three moves, hiding behind a sentence.

17. Error analysis. Jo says the line through (1, 4) perpendicular to $y = 2x + 3$ is $y = 2x + 2$. Name Jo's mistake in one sentence, then write the correct equation.

answer: _____

18. Write the line through (−2, 7) that is parallel to the x-axis. Then write the line through (−2, 7) that is perpendicular to the x-axis.

answer: _____

19. Line a passes through (−1, 2) and (3, k). Line a is perpendicular to $y = −2x + 1$. Find k.

answer: _____

20. Line m is $y = \frac{1}{3}x + 2$. Line n is perpendicular to m and passes through (6, −1). Do m and n cross on the y-axis? Find where they cross and justify.

answer: _____

✓ **Check yourself:** Add your k from #19 to the x-coordinate of the crossing point in #20. You should get 8.5. If not, redo #19 first. It's faster.